

EXHALE: Convergence Reliability and the Steady-State Newton (JFNK) Solver

(work log, 2026-06-10/11)

This memo is the complete record of two tightly linked developments in EXHALE: (i) the discovery that the historical convergence criterion ($du < 10^{-3}$ on the mass-flux spread) can stop runs at premature, non-steady snapshots (the WASP-121b reference used throughout earlier work was such a snapshot, low by ~ 0.08 dex in $\dot{M}$) together with the new convergence diagnostics that fix this; and (ii) the construction, debugging, and first converged run of a direct steady-state solver (pseudo-transient continuation + Jacobian-free Newton–Krylov), which reaches residual levels unattainable by time marching at $\sim 1/200$ of the cost. Every code change is opt-in: with all new options at their defaults, the marching code is byte-identical to the pre-existing behavior (verified by the regression harness after every increment). Statements of physical interpretation below are tentative unless explicitly verified.

***Note added 2026-08-10.** This is a work log of 2026-06-10/11 and is kept as written. Three of the solver constants it records have since been replaced: the diagonal scaling, the stagnation watchdog, and the warm-up hand-off criterion. Section 8. lists what changed and what it means for the stall diagnoses recorded here, in short, the repeated finding that “every stall pins to the base cells” was an artifact of the old scaling and of a worst-cell print that normalized momentum by the base cell’s own $|\rho v|$.*

Contents

1. Background and Stage-I refactor (context)	2
2. The convergence-criterion problem	2
2.1 What du measures, and what it misses	2
2.2 The premature-dip discovery (WASP-121b)	2
2.3 New diagnostics and stops (all opt-in)	3
3. The steady-state solver	3
3.1 Formulation	4
3.2 Jacobian and linear algebra	4
3.3 The five necessary ingredients (each found by a stall)	4
3.4 Result: first converged steady solve	5
4. Runtime options and hooks (reference)	6
5. Hardening results (2026-06-11, after first convergence)	8
6. Remaining open items	9
7. File inventory	10

8. Postscript: what changed after this log (2026-08-10/11)	10
9. The stationary solve as it stands (2026-09)	11
9.1 The merit reads every row on the scale it will be judged by	11
9.2 The trust region, and the linear solve under it	12
9.3 The element constraint rows	13
9.4 The column floor and the equilibrated preconditioner	13
9.5 Three arms that were measured and are OFF	14
9.6 What limits the species-row solves today	14
9.7 The residual contract	15
9.8 What a certified state is now, and what refuses one	16
9.9 Numbers in this section	16

1. Background and Stage-I refactor (context)

The work below sits on top of the Stage-I “extensibility” refactor of the same dates (named species constants; the composition single policy point for n_e/n_{tot} /EOS; schema-2 output headers and a label-mapping `load_IC` that preserves the metal ionization state on restart; element-loop abundances; rate dispatch by ion index). Each step was a no-physics-change refactor gated by a deterministic regression harness (`backup/regression/run_check.sh`: single-threaded WASP-121b runs, golden outputs compared byte-wise excluding # header lines; matrix cases `wasp_full = He 23S + metals` and `wasp_he23off`). Details: `docs/refactor_stage1_log.md`. The same harness gates every solver-related change in this memo.

2. The convergence-criterion problem

2.1 What du measures, and what it misses

ATES declares steady state when

$$du = \frac{\max_j |\rho v r^2|_j - \min_j |\rho v r^2|_j}{\min_j |\rho v r^2|_j} \Big|_{j \in [j_{\min}, N]} < du_{\text{th}}, \quad du_{\text{th}} = 10^{-3}, \quad (1)$$

i.e. the *spatial spread* of the mass flux over the wind window ($r \geq r_{\text{esc}}$). Two blind spots were demonstrated this session:

1. du cannot see a *uniform level drift*: the whole $\rho v r^2$ profile can rise or fall while staying flat.
2. du is *oscillatory* on WASP-121b: it dips below 10^{-3} transiently long before the wind settles.

2.2 The premature-dip discovery (WASP-121b)

The single-thread regression golden (He 2³S + metals, two-stage PLM→WENO3, stopped by $du < 10^{-3}$ at step 7,288) has raw $\log_{10} 4\pi \rho v r^2(r_{\text{max}}) = 13.631$. Suppressing the du stop and marching on, the flux level rises *monotonically* (the windowed level change `lev_rel` falls $0.23 \rightarrow 2.7 \times 10^{-3}$) toward $\log_{10} \dot{M} \simeq 13.71$: at step 35,085 the state reaches

13.707 and is still creeping upward, with $T_{\max}$ falling from 11,021 K to 10,767 K. Three independent paths agree on the deeper state (long cold marching $\rightarrow$ 13.707; warm restarts $\rightarrow$ 13.716; the naive local- dt experiment $\rightarrow$ 13.716 in 1,702 steps). **The old golden was a premature du -dip snapshot, not the steady state:** approximately 0.08 dex ($\sim 20\%$) low in $\dot{M}$.

Two earlier conclusions were corrected as a result. First, naive cell-by-cell pseudo-time stepping (“Time stepping: Local”) had been judged “answer-shifting”; in fact it reaches the *correct deeper* state $\sim 20\times$ faster: the judgment was an artifact of comparing against the premature golden. (The LTS state does carry larger hydro residuals in each cell, §2.3, so it is “fast but needs polishing.”) Second, the apparent “path-dependent quasi-steady states” largely collapse to one true state plus premature snapshots along the way.

2.3 New diagnostics and stops (all opt-in)

Level-stability gate (Level tol:). A converged/stalled stop additionally requires the mean $|\rho v r^2|$ over $[j_{\min}, N]$ to change by less than lev_th (relative) across the last N_{stall} steps (ring buffer in EXHALE_main; the relative level change is now the 4th column of the stdout at each step). This correctly refuses the premature dip (the cold run marches past step 7,288), but a practical tolerance cannot discriminate ultra-slow drifts; it is necessary, not sufficient. Default off.

Steady-residual diagnostic (EXHALE_RESIDUAL=1). The finite-volume steady residual

$$\mathbf{R}_j = \begin{cases} (\nabla \cdot \mathbf{F})_j - S_j, & \text{mass, momentum,} \\ (\nabla \cdot \mathbf{F}_E)_j - S_{E,j} - (\mathcal{H}_j - \mathcal{C}_j), & \text{energy,} \end{cases} \quad (2)$$

evaluated once on a loaded state (one Reconstruct+RK_rhs pass; heating/cooling from ioniz_eq; reported as $\max_j |\mathbf{R}| / \max_j |u|$ per component over the window). This is *independent* of $dudtu/\text{stall}$ and ranks candidate states:

state	R_{mass}	R_{mom}	R_{energy}
golden 13.631 ($du < 10^{-3}$ stop)	9.9×10^{-2}	4.7×10^{-1}	3.1×10^1
warm 13.716 (local- dt + cont.)	2.4	2.8	7.4×10^{-1}
cold 13.707 (35k steps)	2.1×10^{-2}	1.6×10^{-2}	1.5×10^{-2}

The premature golden has an energy residual of ~ 30 : heating and cooling are grossly out of balance, which du cannot detect. (The energy value 1.5×10^{-2} for the cold state corresponds to the consistent ghost-handling order, §3.1.)

Residual-based stop (Resid tol:). In stage 2 the run converges on $\max_k \max_j |\mathbf{R}_{j,k}| / \max_j |u_k| < \text{resid_th}$ (evaluated every $N_{\text{resid}}=500$ steps; cheap), *replacing* the du test. Validation: a run with resid_th= 10^{-2} reached $du = 8.6 \times 10^{-4}$ (below the old stop) at step 32k and correctly kept marching because $R_{\max}$ was still 3.1×10^{-2} and falling monotonically. Default off.

3. The steady-state solver

The marching cost to a genuinely converged wind ($> 35\text{k}$ steps for WASP-121b; $> 10^6$ steps for HD189733b without reaching it) motivates solving the steady residual directly: $F(Y) = 0$.

3.1 Formulation

Unknowns $Y \in \mathbb{R}^{3N}$: the conserved hydro variables $(\rho, \rho v, E)$ on the N physical cells. Ghost cells are not unknowns; Apply_BC fills them from the interior on every residual evaluation (lower boundary: pinned ρ_{bc} , $p = 1 + dp_{bc}$ and the one-way velocity valve; upper: extrapolation). **Local elimination**: given Y , the ionization equilibrium is re-solved (ioniz_eq) and the resulting heating/cooling enter the energy residual, so F is a pure-hydro residual of size $3N$ (“method A” of the SUNDIALS plan), and the non-local radiation (column densities) is inside F . The residual construction is shared with the diagnostics (steady_residual.f90: assemble_residual); the vector form lives in steady_newton.f90 (pack_U/unpack_U/ eval_residual), validated to reproduce the EXHALE_RESIDUAL numbers to all digits. One ordering subtlety: Apply_BC must run *before* the chemistry/heating step so that F depends only on the interior Y ; the diagnostic hook was aligned to the same order.

3.2 Jacobian and linear algebra

Banded approximation. With WENO3’s ± 2 -cell stencil and 3 coupled variables per cell, $\partial F_i / \partial Y_j$ has half-bandwidths $k_l = k_u = 8$ in the flat ordering. A graph coloring with stride $k_l + k_u + 1 = 17$ recovers all band entries of a *local* residual from 17 finite-difference probes. The banded matrix is factored with LAPACK dgbtrf/dgbtrs (LDLIBS=llapack; ATLAS LAPACK 3.7.1 on this machine, link-tested). A unit test (EXHALE_JAC_TEST=1) checks Jr against a directional finite difference: relative mismatch 1.3×10^{-4} , i.e. the band layout is correct (an error would be $O(1)$).

JFNK. Because the radiation makes F weakly non-local, the banded matrix is used as a *preconditioner* while the true Jacobian-vector product is matrix-free, $Jv \simeq [F(Y + \varepsilon v) - F(Y)] / \varepsilon$ (jv_product), inside a right-preconditioned GMRES(m) (pgmres, modified Gram–Schmidt + Givens; inexact-Newton forcing 10^{-1}). The outer loop is pseudo-transient continuation,

$$\left(\frac{1}{\Delta\tau} I + J \right) \delta Y = -F(Y), \quad \Delta\tau \leftarrow \Delta\tau \max(\lambda, 0.1) \frac{\|F\|_{\text{old}}}{\|F\|_{\text{new}}} \quad (\text{SER ramp}), \quad (3)$$

with a backtracking line search and positivity guards ($\rho, p > 0$).

3.3 The five necessary ingredients (each found by a stall)

The path to convergence was a sequence of well-localized stalls; each fix was verified by its own probe (warm start from the cold-35k state, $\Delta\tau_0 = 1$ unless noted). Removing any one ingredient reproduces the corresponding stall.

1. **Smooth line-search merit.** The first driver used the component-wise max-relative residual as the merit; it is non-smooth and rejected every step ($\lambda \rightarrow 10^{-6}$). Fix: merit = $\|F\|_2$ (later $\|D^{-1}F\|_2$). After the fix the banded-PTC driver descends, but only linearly ($\sim 2\%$ /iteration), and a $\Delta\tau_0 = 1$ probe showed a textbook Newton step (residual halved in one step) before stalling, proving the approach and isolating the next blocker.
2. **Matrix-free Jv (JFNK).** The banded Jacobian omits the non-local radiation response; with exact Jv via GMRES the merit fell $4.5\times$ in two steps (vs. 2% /step), $\Delta\tau$ grew. Stall again: now localized (whole-domain worst-cell diagnostic) to the *base* cells $j = 1-2$ (an earlier “sonic point at r_{esc} ” reading was an artifact of searching only $[j_{\text{min}}, N]$; the sonic point is actually at $r = 1.61$ and r_{esc} enters no physics). Freezing base cells with anchor rows does *not* help: the worst cell simply follows the slab edge and the anchor interface itself introduces a kink.

3. **Diagonal scaling.** The unknowns span ~ 7 decades (base $\rho \sim 1$ vs. wind $\sim 10^{-7}$ in code units); FD steps with absolute floors give wind cells 15–60% relative perturbations and corrupt exactly the columns where the residual lives after the first base-scale corrections. With $D_i = \max(|Y_i|, 10^{-6} \max_j |Y^{(k)}|)$ (the 2026-06 form; replaced 2026-08-10, §8.), the scaled system $D^{-1}(\frac{1}{\Delta\tau}I + J)D$, D-relative FD steps, and merit $\|D^{-1}F\|_2$: GMRES needs 2–6 vectors (was 9) and the merit falls $24\times$ before stalling, at the base again.
4. **Smooth base valve.** The lower BC ghost velocity is $\max(v_1, 0)$, and the steady state hugs the kink ($v_1 \simeq +4.5 \times 10^{-3} c_s$): Newton trial steps crossing $v_1 = 0$ are non-differentiable. Opt-in replacement (Valve eps): $v_g = \frac{1}{2}(v + \sqrt{v^2 + \varepsilon_v^2})$, smooth, $\rightarrow 0$ for $v \ll -\varepsilon_v$, $\rightarrow v$ for $v \gg \varepsilon_v$ (an earlier variant with a $-\varepsilon_v/2$ shift leaked inflow and was fixed). With $\varepsilon_v = 10^{-2}$ (too large, it perturbs the state) the solver descends $10\times$ through the base (differentiability is what Newton needed), but $\varepsilon_v = 10^{-4}$ still stalled at the base: the valve was not the only kink.
5. **Frozen WENO weights + non-monotone line search.** The ESWENO3 smoothness factors $S_{0,1} = 1 + \tau^2/\beta_{0,1}$ are smooth but strongly nonlinear; freezing them at the outer iterate (weno_mode: store on a mode-1 pass, reuse on mode-2 passes for all Jacobian probes, Jv products, and line-search trials, the standard lagged-weights remedy) removes their response from the inner problem. Independently, the monotone Armijo test fails once the required decrease $10^{-4}\lambda\|F\|$ falls below the iterative-chemistry noise floor of the residual; a *non-monotone* (Grippo, memory 5) acceptance (improve on the worst of the last five accepted merits) is the standard remedy.

3.4 Result: first converged steady solve

With all five ingredients (warm start from the cold-35k state, $\Delta\tau_0 = 1$, $\varepsilon_v = 10^{-4}$):

target	iterations	final $\ R\ $
Resid tol = 10^{-3}	8 (all $\lambda = 1$, GMRES 2–3)	9.5×10^{-4}
Resid tol = 10^{-4}	22	6.0×10^{-5}

Each outer iteration costs ~ 25 residual evaluations (17 for the banded preconditioner + GMRES + line search), so the 10^{-3} solve costs the equivalent of ~ 200 marching steps, versus $> 35,000$ marching steps that never reached this residual level. Convergence is linear, not quadratic (the frozen weights are refreshed per outer iteration (a Picard–Newton hybrid), and the chemistry noise floor remains); every step is a full Newton step.

Physical validation (vs. the marching references):

state	$\log_{10} \dot{M}_{\text{raw}}$	spread($r \geq 1.5$)	T_{max}
Newton (8 its)	13.7111	5.0×10^{-4}	10,747 K
cold marching (35k steps)	13.7066 (rising)	1.2×10^{-3}	10,767 K
old golden (premature)	13.6309	1.8×10^{-3}	11,021 K

The Newton solution sits on the marching asymptote ($\simeq 13.71$), is the flattest state produced to date, and agrees with the deep-marching reference to 0.005 dex in $\dot{M}$ and 0.2% in T_{max} .

4. Runtime options and hooks (reference)

All defaults preserve the legacy behavior byte-identically (re-verified by the harness after every change in this memo).

input line	default	effect
du_th [PLM,WENO3]: a b	-	convergence thresholds; the single-stage vs. two-stage choice is set by Reconstruction scheme: (PLM/WENO3 single-stage use only a ; PLM+WENO3 two-stage: PLM until $du < a$ or stall, then WENO3 until $du < b$)
Stall [tol,N]: t n	10^{-6} , 2000	stall-detector override
Level tol: v	off	mass-flux level-stability gate (§2.3)
Resid tol: v	off	residual-based convergence (§2.3)
Time stepping: Local	off	cell-by-cell pseudo- dt (steady-state acceleration; reaches the deep state fast but leaves residuals in each cell)
Energy solver: Explicit	semi-implicit	original forward-Euler source update (isolation tests)
Valve eps: v	off (hard $\max(v, 0)$)	smooth base valve (§3.3)
CFL: v	0.6	CFL override

environment hook	effect
EXHALE_DUMP_IC=1	write the loaded state and stop (restart round-trip test)
EXHALE_RESIDUAL=1	evaluate and report the steady residual of the loaded state, then stop
EXHALE_NEWTON_TEST=1	pack/unpack + vector-residual self-test
EXHALE_JAC_TEST=1	banded-Jacobian vs. directional-FD unit test
EXHALE_PTC=1	run the steady solver from the (loaded) IC, write output, stop; EXHALE_PTC_JFNK=1 selects JFNK (else banded PTC); EXHALE_PTC_DTAU0, EXHALE_PTC_NFIX optional
<i>added with the species-row route, §9.</i>	
EXHALE_JFNK_MAXIT= <i>n</i>	cap the outer iterations of a stationary solve at <i>n</i> in place of the caller's number. It also caps the RUN: under it the run takes its FIRST stationary solve and refuses later entries by name, so one measurement is one solve. Without it the cap is jfnk_outer_iterations_default = 3000 at all three call sites
EXHALE_INVENTORY_REPORT=1	print the elemental inventory of every cell at the carrier operator's entry (§9.3)
EXHALE_ELEMENT_CONSTRAINT_ROWS=0	restore the five coordinate-box corners in place of the shared element rows
EXHALE_ELEMENT_WRITE_BACK	the element-conserving write-back of each iterate
EXHALE_COLUMN_EQUIL=1	equilibrate the columns of the MODEL as well as of the factored band; measured and not adopted (§9.4)
EXHALE_TR_RESTART_STALL= <i>n</i>	restart the trust-region state after <i>n</i> outer iterations without improvement; default disarmed (§9.5)
EXHALE_KRYLOV_ON_THE_BALL=1	Steihaug–Toint truncation of the Krylov cycle at the radius; default off (§9.5)
EXHALE_GM_REORTHO=1	reorthogonalize the Arnoldi basis; default off (§9.5)
EXHALE_CAUCHY_LEG_BY_IMAGE=1	let the image-share test alone decide the approximate-gradient leg; worse on both fixtures, default off
EXHALE_CERT_ANCHOR=1	run the five tolerance anchors of §9.8 and print them; the default certification report is byte-identical without it
EXHALE_ELEM_DIAG	the diagnostic line the species-row step prints at every trial

Working recipe for the converged WASP-121b solve:

```
# input.inp: ... + "Valve eps: 1.0e-4" + "Resid tol: 1.0e-3"
# output/{Hydro_ioniz,Ion_species}_IC.txt: a warm-start state
EXHALE_PTC=1 EXHALE_PTC_JFNK=1 EXHALE_PTC_DTAU0=1.0 ./EXHALE.x
```

5. Hardening results (2026-06-11, after first convergence)

HD189733b: 33 s vs. >3 h. Warm-started from the deepest marching state available (1.16M steps, $du = 6.6 \times 10^{-3}$; starting residual $(2.4, 5.2, 2.4) \times 10^{-3}$), the JFNK solver converged to $\|R\| = 9.5 \times 10^{-4}$ in 129 iterations / **33 seconds wall time**. The converged state has $\log_{10} \dot{M}_{\text{raw}} = 10.298$ vs. 10.218 (marching) and 10.223 (pristine ATES-Code-main) (both still drifting upward), and is the flattest state produced (spread 7.6×10^{-3} vs. 1.25×10^{-2}). Tentatively, the true HD189733b $\dot{M}$ is $\sim 20\%$ above the marching estimates, the same premature-snapshot pattern quantified for WASP-121b. Tightening to 10^{-4} grinds: the frozen-weight refresh jumps are large at the breathing base and progress floors at $\|R\| \sim 3.4 \times 10^{-4}$, the practical floor for this planet with the current settings.

Valve- ϵ insensitivity. Two full Newton solves of WASP-121b (tol 10^{-4}) with $\epsilon_v = 10^{-4}$ and 10^{-5} give $\log_{10} \dot{M} = 13.7110$ and $T_{\text{max}} = 10750.2$ K *identically* to the digits quoted: the valve smoothing is a pure numerical regularization.

Solver: Newton (production wiring) and the switch criterion. The warm-up $\rightarrow$ Newton hand-off was wired into the main loop. Two failure modes were found and fixed along the way, both reiterating the central lesson: (i) a du -based switch fails: at $du < 10^{-2}$ during the early transient the residual is still $\|R\| \approx 4.8$, far outside Newton's basin; (ii) with the switch fixed, the marching's own $du < 10^{-3}$ stop fired first, returning the premature state. The design adopted at the time: *the switch and the warm-up termination are both residual-based* (periodic monitor; switch at $\max \|R\| < 5 \times 10^{-2}$, du -stops suspended in this mode). *This was later reverted to a flux-keyed hand-off; see §8.* End-to-end validation from a *cold isothermal IC* (WASP-121b, single input line Solver: Newton): two-stage warm-up to step 27,000 ($\|R\| = 4.85 \times 10^{-2}$) $\rightarrow$ JFNK 16 iterations $\rightarrow \|R\| = 5.4 \times 10^{-4} \rightarrow$ standard outputs and post-processing; final $\log_{10} \dot{M}_{\text{raw}} = 13.7111$, spread 5.1×10^{-4} , $T_{\text{max}} = 10,748$ K, identical to the warm-started reference. Wall time 16 min (4 threads), dominated by the warm-up.

Transmission spectra of the Newton solutions. EXHALE_transit.py was run on both converged solutions (after PP-only runs generated the $\star_{\text{adv}}$ profiles): He 10830 / Ly α / H α / H β spectra for both planets plus the metal lines for WASP-121b (Mg II 2796: 3.7%, Ca II K: 3.8%, Na I D2: 1.7% line-center). Peak line-center absorptions: WASP-121b He 3.7%, H α 3.7%; HD189733b He 11.9%, Ly α 46%, H α 1.8% (consistent in magnitude with the Jensen et al. 2012 H α detection). Figures and numbers are collected in backup/regression/steady_solver_results.ipynb; usage is documented in the updated user manual (§ EXHALE_transit.py).

Failure containment (a genuine JFNK failure, and two fixes). Re-running the manual's four figure runs through the production pipeline exposed a real failure mode on the spherical, metals-off tutorial planet (a generic hot Jupiter, domain to $10R_p$). The solve nearly converged ($\|R\| = 6.0 \times 10^{-3}$ at iteration 2), but the non-monotone reference still held the huge *initial* merit ($\|F_s\|_2 \simeq 10^4$), so a blow-up step was “accepted”; the search then stalled at $\lambda \simeq 10^{-6}$, accepting zero-length steps against the now-flat reference for ~ 470 iterations, and (worst of all) the main loop accepted the final unconverged state ($\|R\| = 0.42$) as converged and wrote it out. Two fixes (both in the default-off Newton path; legacy byte-identity re-verified): (i) *best-iterate tracking and fail-fast* in solve_steady_jfnk: the best residual state is stored, and the solve aborts with info=2 after 15 outer iterations without a new best, returning the best iterate (*this rule was*

replaced 2026-08-10, §8.); (ii) in EXHALE_main, `info≠0` is treated as a failure: the run prints `JFNK failed`, keeps the best iterate, disables the Newton mode, and *resumes plain time-marching* with the *du*-based stops re-armed, so an unconverged Newton state can never masquerade as the final answer. Re-run with the fixes, the same configuration aborts after 15 stuck iterations, returns the best iterate ($\|R\| = 5.97 \times 10^{-3}$), and the resumed marching stops one step later ($du = 3.4 \times 10^{-4}$, already below threshold): the run ends on an honest, near-converged state instead of a diverged one. The three other figure runs converged cleanly through the pipeline (tutorial metals-on $\|R\| = 5.6 \times 10^{-4}$, WASP-121b spherical Case A 4.1×10^{-4} , Roche Case B 5.8×10^{-4} , all `info=0`).

Can JFNK start cold? (basin-boundary experiments, 2026-06-11). Three probes on WASP-121b (all with the fail-fast guard, so each failure is cheap and the run still finishes honestly via the marching fallback):

start of JFNK	start $\ R\ $	JFNK outcome	final answer
cold isothermal IC	1.4×10^4	1 iter $\rightarrow$ 11, hard stall at the base	- (abort)
step 6,500 (<code>R_switch=0.5</code>)	0.43	$\rightarrow$ 0.245, stall	fallback <i>du</i> -stop, $\dot{M}$ 13.34 (premature)
step 12,500 (<code>R_switch=0.1</code>)	0.091	first step <i>worsens</i> (0.148), stall	fallback <i>du</i> -stop, $\dot{M}$ 13.38 (premature)
step 27,000 (default 5×10^{-2})	0.0485	16 iters $\rightarrow$ 5.4×10^{-4}	$\dot{M}$ 13.7111 ✓

The Newton basin boundary on this planet therefore lies between $\|R\| \simeq 0.09$ and 0.049, the default switch is calibrated essentially at the edge, and raising it buys nothing: every stall pins to the first base cells ($j = 1-2$, momentum), where the valve/ionization coupling makes the frozen-weight local model invalid far from the fixed point. PTC's small- $\Delta\tau$ limit is just implicit time stepping at $\sim 20-60\times$ the cost per unit pseudo-time of an explicit RK step, so a cold-start PTC could not beat the marching warm-up economically even where it survives. Practical caveat: when a Solver: Newton run prints `JFNK failed`, the fallback finishes on the *du* stop, which can be a premature state (13.34/13.38 vs. 13.71 here): re-check such runs with `EXHALE_RESIDUAL=1`. Probes preserved in `backup/regression/{jfnk_cold, newton_rsw05, newton_rsw01}`.

6. Remaining open items

- HD189733b residual floor.** Progress floors at $\|R\| \sim 3.4 \times 10^{-4}$ (frozen-weight refresh jumps at the breathing base); an under-relaxed weight refresh between outer iterations is the natural next step. *Since addressed from the other side: the line search no longer accepts a step on the frozen-weight measure at all, item 4 of §8.*
- Non-monotone reference poisoning.** The tutorial-planet failure above was *contained*, not prevented: the 5-entry Grippo memory can retain a huge early merit and thereby accept a blow-up step. Capping the reference at a multiple of the current merit (or flushing the history after a large drop) should prevent it outright.
- Chemistry noise floor.** The non-monotone search works around the `ioniz_eq` iteration noise; tightening the inner solve (or smoothing its switch points) would restore certifiable descent and possibly quadratic tails.
- Quasi-steady physics.** The deep states still show slow residual decay rather than an exact fixed point (base breathing); whether an exact steady state of the discrete scheme exists with the hard valve remains open: the smooth-valve fixed point found here is the natural regularized answer.

These four are the open items of the three-unknown solve, as it stood in 2026-06. The species-row route opened its own, and they are stated where they arose: the linear model and its Arnoldi gap (§9.6), the column preconditioner as the untried freedom of the same section, the conditional closure convergence left FAIL by design (§9.7), and the layer's element flux conservation, which is what keeps the certification from gating below $r = 1.20$ (§9.8). No species-row configuration converges today, and none is certified.

7. File inventory

file	content
src/modules/time_step/steady_residual.f90	shared steady-residual assembly + norms (diagnostics, monitor, Newton)
src/modules/time_step/steady_newton.f90	pack/unpack, residuals, scaling, banded Jacobians (frozen/full), jv_product, pgmres, solve_steady_ptc, solve_steady_jfnk, base-anchor option
src/modules/states/Reconstruction.f90	weno_mode frozen-weights support (S0/S1 store/reuse)
src/modules/states/Apply_BC.f90	smooth base valve (valve_eps)
src/modules/init/parameters.f90	new option variables (defaults off)
src/modules/files_IO/input_read.f90	option parsing (keyword lines after Force start:)
src/EXHALE_main.f90	level/residual monitors and stops; env hooks
backup/regression/	harness, goldens, and all probe run directories (ptc_warm, crit_cold, resid_*, ...)
docs/steady_solver_design.md	incremental design notes (§§4–11) with the raw probe logs
docs/refactor_stage1_log.md	Stage-I extensibility change log

8. Postscript: what changed after this log (2026-08-10/11)

Four things recorded above are no longer what the code does.

1. **The hand-off is flux-keyed, not residual-keyed.** The Solver: Newton key hands off when the flux metric du (the radial spread of pvr^2) falls below `newton_du_switch`, default 10^{-2} , settable as Solver: Newton <du_switch>. The R_switch keys of the tables above no longer exist.
2. **The diagonal scaling is local and parameter-free.** `build_scaling` was replaced by `cell_state_scales`, which builds every scale from the cell's own state: ρ , $\rho(|v| + c_s)$, E . The old momentum floor $10^{-6} \max_j |\rho v|$ was set by the base cell, which holds the global maximum of $|\rho v|$ while carrying no wind, so cells where the flow reverses were divided by a number unrelated to their own state.
3. **The stagnation watchdog counts failed line searches.** It aborts after 12 consecutive outer iterations in which the line search found no acceptable step, instead of after 15 iterations without a new best $\|R\|$. Every solve measured needs 27–38 iterations before it first improves on the warm-start $\|R\|$, so the old rule fired during the opening pseudo-transient in normal operation.

4. **The line search is tested against the true residual** (2026-08-11). The trial states of the backtracking search in `solve_steady_jfnk` are evaluated with `weno_mode = 0`, i.e. with the WENO smoothness weights recomputed at the trial state, so what decides acceptance is the residual the solve is driving to zero rather than the frozen-weight model of the current outer iteration. On the HD 189733 b hand-off state the two measures differ by a median factor 2.2 at $\lambda = 1$, and 7 of the 14 steps the frozen test accepted *raised* the true residual, by up to $9\times$. This is the change that attacks the HD 189733 b residual floor of item 1 of §6. from the other side. Numbers: `docs/newton_scaling_and_base_wall.md` §10.

The consequence for this memo is that its recurring diagnosis (that every stall lands on the base cells $j = 1-2$, momentum) does not survive re-measurement. The base momentum row is satisfiable: its residual is the remainder of a four-order cancellation of gravity against the pressure gradient, 6.5×10^{-5} of the gravity term, and a 2.9 ppm change of the ghost pressure drives it to zero. What actually dominated the line-search merit were the momentum rows around the stagnation point, mis-scaled by the floor above. Configurations that this memo records as stalling now converge with `info=0`. Full account and numbers: `docs/newton_scaling_and_base_wall.md`.

9. The stationary solve as it stands (2026-09)

The solver of §3. solved three unknowns per cell. Since 2026-09 it also solves the SPECIES ROWS: one row per transported element and one per molecular carrier, four to eleven unknowns per cell depending on the configuration. Everything in this section is that route; the three-unknown route is untouched by construction and every configuration that used it, `wasp_full_newton` included, is bitwise across the whole series.

This section replaces the description of the outer loop in §3.1–§3.3 for the species-row route: the backtracking line search of the pseudo-transient continuation is no longer what decides the step there. It is a trust region.

9.1 The merit reads every row on the scale it will be judged by

There is one denominator for a species row, `merit_row_scale_from_certification`, and the acceptance test, the merit of the line search and the certification all divide by it. The hydrodynamic rows keep `residual_row_scale(k, j, u)`, the element rows take their certification scale, the carrier rows take the term sum the certification divides by, and the base cells of element rows keep the unknown's own scale, since the base is the Dirichlet reservoir and carries no equation. MEASURED on both species-row fixtures: every element row's largest scaled merit row now EQUALS its certification measure at the same cell, where before the two stood a factor 2.5 apart at the first iteration and 112 apart at the sixtieth.

The measured consequence was not free, and is recorded rather than hidden: reconciling the hydrodynamic rows onto the same footing improved the carrier fixture by $2.6\times$ and made the atomic element fixture stop solving altogether, because the certification row scales span 11.4 decades against 8.3 before. The cause turned out not to be the spread but one exactly empty Jacobian column (§9.4).

9.2 The trust region, and the linear solve under it

The step is a dogleg inside a trust region on the scaled coordinates, and every part of it now has a name that is printed.

Named linear outcomes. `pgmres` returns one of seven:

`gm_tolerance_reached`, `gm_subspace_exhausted`, `gm_no_direction_sampled`, `gm_jacobian_action_unusable`,
`gm_preconditioner_failed`, `gm_reduced_system_singular`, `gm_nonfinite`.

A zero right-hand side returns before any factorization; the rotations use hypot; the rank of the reduced system is tested before the triangular division and the leading nonsingular block is the one solved; the LAPACK status and finiteness are checked after every preconditioner apply; at a breakdown the true scaled residual is verified with one extra product and the cycle exits.

The radius is initialized from a direction that exists, and has a floor. `initial_trust_region_radius` takes its first radius from a finite Krylov direction with the active bounds held, and the floor $\text{tr_delta_min} = 10^{-6} \sqrt{n_{\text{eq}}}$ in scaled coordinates stops a radius of exactly zero from being an absorbing state; the ceiling `tr_dmax` follows the radius and cannot pass $10^3 \sqrt{n_{\text{eq}}}$. “No feasible descent” is `info = 3`.

This is the fix of a real failure, and its mechanism is worth recording because it looked like a solver defect and was a registration defect: on the atomic element reload 1500 unknowns sat on a bound and the first cycle took zero products, the radius became zero and stayed there. The 1500 were the three elements `metals.inp` does not carry (Si, K, S over 500 cells), registered with an identity row and sitting exactly at zero, so the fraction-to-the-boundary rule annihilated any direction with a component there. Absent elements, those with reservoir abundance ≤ 0 , are no longer registered at all. MEASURED: unknowns on a bound $1500 \rightarrow 0$, first radius $0 \rightarrow 1.909 \times 10^{-2}$, first step accepted at ratio 0.995.

The length along the direction used, and the leg that is dropped. The step length along the approximate-gradient leg is $\tau = (r \cdot Ag) / \|Ag\|^2$, the interior minimizer along the leg the step is actually taken on. That leg is DROPPED from the step and from its model image on two tests: by length, when it is shorter than `cauchy_leg_min_fraction` $= 10^{-3}$ of the radius, and by image share, when it supplies more than `cauchy_leg_image_share_max` $= 0.5$ of the model image while being less than `cauchy_leg_step_share_max` $= 0.1$ of the step. It is dropped and the dogleg rebuilt at no extra product, not rescaled: rescaling would either shorten As_U without s_U , which is not a step of the operator, or move the step off the dogleg path. The reason for the second test is that the first one is a proxy: a leg that is 0.7% of the step can supply ALL of its image. Honest impact: on the fixtures as they now run, the length test reaches every image-dominant leg first (117 length drops, 0 image drops), so the image rule fires zero times and all three runs are bitwise. The rule is the right criterion nonetheless and is kept; it is not claimed to improve either fixture.

The ray test has a measured floor and a sound-ratio rule. The decision is `ray_slope_verdict`: the model slope along the step is compared with the slope measured along the ray, and the floor under the comparison is $\max(\text{eq_sweep_reltol} |f_2|, \text{ray_floor_factor} \times \text{measured reproducibility})$, the reproducibility measured once per outer iteration by two evaluations of the merit at one state (cost: one residual evaluation per outer iteration, +1.5% on the carrier fixture). MEASURED, the measurement refutes what the diagnostic before it claimed: the reproducibility is

1.8×10^{-15} at a merit of 14.95 and 0 to 4×10^{-19} at 2.7×10^{-5} , four to seven decades below the assumed part, which therefore governs everywhere, and the refused trials' $|f_2^+ - f_2^-|$ stood two to three decades above the floor, so they were real slope differences and not a probe at its cancellation floor. Within `ratio_sound_band` = 10^{-2} of the acceptance ratio the ray verdict is formed and printed but not acted on.

The face-write hold. A trial whose species unknown would be written onto a face of its box is a move of the state that no model of the step contains, and those were exactly the trials whose predicted decrease stopped scaling with the step below $\|s\| \sim 10^{-7}$ (MEASURED: 11 of 11 such trials refused at a fixed ascent of 3×10^{-3} over five decades of step; 10 of 10 others accepted at ratio 1.000). `hold_the_step_where_it_leaves_the_species_box` zeroes the step at such an unknown, so the trial stays where the operator can be sampled. `probe_length_of_the_jacobian_action` is now the one definition of the probe arc. Atomic fixture at a cap of 300, before $\rightarrow$ after: 227 $\rightarrow$ 113 iterations, 170/55 $\rightarrow$ 91/19 accepted/refused, negative ratios 49 $\rightarrow$ 14, best $\|R\|$ $2.353 \times 10^{-2} \rightarrow 1.333 \times 10^{-2}$; the period-two cycle is gone. The carrier fixture and `wasp_full_newton` are bitwise (the branch never fires there).

9.3 The element constraint rows

The coordinate box that used to limit the species step had one side per carrier, which is the wrong feasible set by stoichiometry: two carriers of one element each spend the whole budget, and a cell at both of its hydrogen-bearing ceilings breaches the shared hydrogen budget by exactly 1.0 of the available hydrogen. The step is now limited by the shared rows themselves, $2n(\text{H}_2) + n(\text{H}^+) + n(\text{OH}) + 2n(\text{H}_2\text{O}) \leq n_{\text{H}}^{\text{avail}}$ and its oxygen and carbon counterparts, each of the old corners being implied by the row, with a nonlinear feasibility screen by magnitude at every candidate and a restoration phase for an infeasible start. The derivatives of the constraint come from the same map that states it (`element_constraint_derivatives` of `lower_atmosphere/element_inventory.f90`).

The write-back of each iterate returns the element totals it was handed. That is where the improvement came from, and it is worth separating from the constraint itself: on the carrier fixture the He/H nucleus ratio stays at 1.39×10^{-11} of its reservoir at every one of 68 iterates where the old text drifted to 2.016×10^{-1} over 79; best $\|R\|$ $0.1513 \rightarrow 0.0899$ at 14% fewer residual evaluations. **Stated, not claimed:** on both fixtures the shared hydrogen row is never within a probe step of being tight, so the active-set PROJECTION onto the constraint hyperplane, the part the route is named for, fired on ZERO iterations of either run; its correctness rests on the suite rows, and the measured gain comes from the write-back, from the budget leaving the coordinate box, and from the shared step limit. Switches: `EXHALE_ELEMENT_CONSTRAINT_ROWS=0` restores the five corners, `EXHALE_ELEMENT_WRITE_BACK` the write-back.

9.4 The column floor and the equilibrated preconditioner

The collapse of §9.1 was one exactly empty Jacobian column. At iteration 11 of the atomic fixture the carbon unknown of cell 500 had been driven to zero, its column scale sat on the absolute floor 10^{-20} , the probe step $\sqrt{\epsilon} \times 10^{-20}$ underflowed the residual difference, the column was exactly zero and the banded preconditioner singular (dgbcon 3.98×10^{-27}). Two repairs, with the model left untouched to the last bit ($r_0 = F/D_{\text{row}}$ and A unchanged, verified by the scaled column norms):

1. `element_unknown_column_scale` = $\max(|x|, 10^{-4}|\text{reservoir}|, 10^{-20})$, so the column stays resolvable while the probe step stands far above $\epsilon X_{\text{reservoir}}$;
2. `equilibrate_the_preconditioner`, a two-sided equilibration in powers of two of the band that is FACTORIZED only, `banded_preconditioner_solve` applying $C \text{dgbtrs}(Rv)$.

Attribution on the first solve (accepted/refused; empty columns at iteration 11; `dgbcon` there): entry 14/16, $1, 4 \times 10^{-27}$; floor alone 69/16, 0, 4.9×10^{-11} ; both 114/36, 0, 1.1×10^{-9} . With a cap of 150 the atomic fixture goes from an abort at $\|R\| = 1.777$ to 3.800×10^{-2} still descending, and the carrier fixture from 0.446 to 0.1513. The model's OWN column equilibration (`EXHALE_COLUMN_EQUIL=1`) was built, measured and NOT adopted: it moves the trust-region coordinates, the radius runs to the cap and trials ascend by 4×10^4 times the predicted decrease.

Two further pieces of the same work: the first-cell rows are declared once (`species_row_base_equation`: an element row is the Dirichlet reservoir, a carrier row the control-volume balance of the operator's own $j = 1$ branch), and the tolerance the composition elimination is asked for is separated from what the outer residual may keep (`eq_sweep_reltol` = `outer_residual_closure_target` divided by the measured amplification of the closure map at the base).

9.5 Three arms that were measured and are OFF

Each of these was implemented and measured against both fixtures, and each is delivered disarmed because the measurement did not support it.

arm	switch	why it is off
periodic restart of the trust-region state	<code>EXHALE_TR_RESTART_STALL</code> (<code>tr_restart_stall_default = 0</code>)	no reset helps: atomic 2.671×10^{-2} either way, carrier $0.090 \rightarrow 0.115$
Steihaug–Toint truncation on the ball	<code>EXHALE_KRYLOV_ON_THE BALL</code>	cheaper (products $33 \rightarrow 7, 40 \rightarrow 1.5$) but carrier $0.090 \rightarrow 0.486$
reorthogonalization of the Arnoldi basis	<code>EXHALE_GM_REORTHO</code>	orthogonality loss is 10^{-13} to 3×10^{-12} ; worse on both: 1.487 and 0.43

The restart measurement also corrected a reading of the run's own behavior that had been taken for a solver property: a run whose first stationary solve returns `info = 1` does not re-enter the solve from the handed-back state, it returns to MARCHING for about 2000 steps and the second solve enters from the marched state. The two decades that a 150+150 route appeared to gain over a single 300 are gained by the marching in between.

9.6 What limits the species-row solves today

Not the step control, and this is measured rather than argued. At the binding iterate of the atomic fixture: zero unknowns held, zero refused samples, zero empty columns, `dgbcon` 1.25×10^{-6} , step ratio 0.9998, merit reproducibility exactly zero, the trust region inert. What fails is the linear model. The cycle spends 40 of 40 products on a direction whose true relative residual is 0.58 against the 0.1 asked (and 0.9917 at the later binding iterate of the same run), with the Arnoldi image 21% away from the operator, and $\|s_N\| = 8.6 \times 10^2$ against a radius 5.9×10^{-2} , so the region uses 7×10^{-5} of an inaccurate direction.

That 21% gap is NOT a loss of orthogonality and not a stale cache. Shown positively: every one of the 40 Arnoldi columns matches its own product to $1-3 \times 10^{-16}$, two products of one direction agree exactly, the basis's orthogonality loss is 4.4×10^{-13} , and yet the action of $V_1 + V_2$ differs from the sum of the two actions by 1.1×10^{-4} . It is the NON-LINEARITY of the finite-difference action across directions, a curvature term over the fixed probe arc $\sqrt{\epsilon}(1 + \|Y\|)$. (One later measurement does not fit a gap that is a function of the assembled length alone, and re-measuring it at three iterates is an open item.)

Doubling the products halves the linear residual and makes the outer solve WORSE on both fixtures, so more of the same direction is not the answer either. The untried freedom is the COLUMN preconditioner; the row scaling is fixed by §9.1.

And the arm is chaotic at the level of the last bit. The atomic element fixture is not a reproducible quantity. A control that multiplies three ghost quantities by $1 + 2 \times 10^{-16}$, changing no equation and posing no boundary, moves $\|R\|$ at iteration 40 from 3.212×10^{-2} to 1.125; a physically correct refill of the same ghosts moves it to 1.199; the three arms part at outer 10–11 and from each other. The amplifier is the Krylov leg at its one-digit tolerance: the relative residual reached differs by 0.3% at outer 10, the promise by 0.4%, then $2.7 \times$ at outer 11, and at outer 12 the three take different branches of the acceptance (0.874 accepted, -1.98 refused, 85 accepted).

The consequence is a rule for anyone measuring this route: “it reproduces the previous arm” rejects correct changes as readily as wrong ones, and $\|R\|$ at iteration N is not what an arm can carry. What it can carry is named outcomes: which row binds, the refusal counts by name, the flux spread, the mass closure.

9.7 The residual contract

Three statements the residual is now held to, and the state of each.

1. **Determinism.** $F(Y)$ is a function of Y , so an evaluation at a point the caller discards (a Jacobian column, a Krylov product, a point on a trust-region ray, a self-test) must put back what it left in the modules it called. The replay registers the species rows and returns 0 of 1500/2000/2500/5500 entries differing (3, 4, 5 and 11 unknowns per cell) after three trials, a discarded trust-region trial and a `ju_product` probe with the active bounds held. GREEN on four configurations. The one part not reachable from the driver is a rejected macrostep. This closed a hole in the old determinism suite, which replayed the THREE-unknown residual in every configuration and so compared 0 of 1500 entries.
2. **Conditional closure convergence.** FAILS, by design, and the failure is the standing measurement. The judged row maximum moves, per unit relative change of the eliminated partition's seed, by 4.2×10^2 carrier tolerances on the carrier fixture and 3.1×10^1 element tolerances on the atomic one; a seed above 2.4×10^{-3} (carrier) or 3.3×10^{-2} (atomic) moves the judged row past its tolerance. Two suite rows are left FAIL rather than given a tolerance that would pass them.
3. **Branch discovery.** With H_2 transported the two-root H_2 front does not reproduce; the $H I$ partition shows one root on the carrier fixture and two on the element-and-carrier fixture, 9 seeds and 1, 1.05×10^{-1} apart, and whether the second is a distinct branch or a slow iteration is open.

The rounding floor of the species rows in the judged measure is exactly ϵ (2.2×10^{-16}) against tolerances of 10^{-8} , 7.65 decades apart, at condition numbers 2 to 10^3 : compensated summation is not a lever here.

9.8 What a certified state is now, and what refuses one

The stationary certification no longer has one tolerance for the species rows. `cert_tol_element_wind = cert_tol_carrier_wind =  $10^{-5}$` , GATING for cells at $r \geq \text{cert_regime_wind_r} = 1.20$; below that the accessors return `cert_tol_reported_only = huge`, so the row is measured, printed and does not gate. One reduction, `certification_species_row_gate`, decides a species row over the gated cells, and a column with no gated cell is refused, never reported as a satisfied row. The report prints gated at $r \geq 1.2$ against $1e-5$: `<measure>` at cell `<j>`, and a certified state with species rows carries the one line saying it is certified IN THE WIND with the rows below it reported; the state file says `cert_reason=certified_in_wind`. The hydrodynamic tolerances (3×10^{-12} , 10^{-8} , 10^{-6}) and the level and closure tolerances did not move. There is no input key: the two radii are properties of the flow and are constants of certification.f90.

The band $1.10 \leq r < 1.20$ is reported and does not gate. The decision was silent on it, and the band holds the candidates' binding cell at $r = 1.153$, where the discretization error of the discrete equation itself is 4.6×10^{-4} : a 10^{-5} gate there would ask the residual to fall 1.5 decades below the error of its own equation, and no anchor for the band exists.

What anchors 10^{-5} , and what refuses 10^{-8} . Five measurements were taken on the two candidate states. The arithmetic does not refuse 10^{-8} : two evaluations of one state differ by exactly zero, the closure-seed floor is 5×10^{-19} , five to eleven decades below it. What refuses it is the DISCRETIZATION and the ELEMENT FLUX CONSERVATION. The derivative of the residual has a clean V at the base (7.6×10^{-8}) but a PLATEAU at 6.6×10^{-5} over four decades of probe length at the binding cell, so the smallest row a Newton step resolves there is 3.9×10^{-8} (6.5×10^{-9} in the wind). A manufactured relaxed column converges at order 1.589 and extrapolates to 1.52×10^{-6} at the production spacing. Grid refinement of the candidate at $r = 1.1515$ takes the He/H row from 5.93×10^{-4} ($N = 500$) to 2.47×10^{-4} ($N = 1000$) with a remap control of 1.5×10^{-5} : about 78% of the row is discretization. And the element flux spread is 9.79 in the layer against 2.7×10^{-2} in the wind. The gate is set a decade above the manufactured column's discretization at the production spacing, and the layer's rows do not gate until the layer's flux conservation is repaired. Full account: docs/certification_tolerance_anchoring_20260910.md.

No candidate is certified. The atomic reload binds on the He/H row of a WIND cell at 2.896×10^{-4} against 10^{-5} ; the carrier reload on the H_2 row at 7.284×10^{-2} . `wasp_full_newton`, which carries no species row, is certified as before.

9.9 Numbers in this section

Every number above is either MEASURED in the item that produced it or READ from the stage-2 record, docs/Update_EXHALE.md section 7, items N0 through N30, and docs/ISSUES_20260909.md. None of them was re-measured for this memo; the fixtures are the two reload directories under backup/regression/ that those items name, and the arm's own irreproducibility (§9.6) is the reason a re-measurement would not reproduce $\|R\|$ at a fixed iteration anyway.